Ayush Agrawal

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Summary

Software engineer with hands-on experience in API and UI test automation, AI-assisted developer tooling, and React/TypeScript frontend development. Built contract-driven test generation frameworks (Python, pytest) and prompt-driven Playwright pipelines, with reporting and CI-friendly workflows. Also experienced in building data pipelines on Microsoft Fabric (DP-600 certified).

Skills

Languages: TypeScript, JavaScript, Python, Java, SQL
Frontend: React, Material UI, HTML, CSS
Testing & Automation: Playwright, pytest, API Testing, Swagger/OpenAPI, Allure, Faker
AI Tooling: OpenAI SDK, Codex CLI, Playwright MCP, Prompt-driven Code Generation
Tools & Cloud: Git, Node.js, Maven, VS Code, Microsoft Azure, Microsoft Fabric

Experience

Software Engineering Intern **Jan 2026 - July 2026**
IALTruist LLC, Remote *Tools Used: Python, pytest, TypeScript, Playwright*

- Developed an **AI-assisted UI test generation POC** using Playwright, TypeScript, and Codex/MCP that converts plain-language prompts into test specs, with a hash-based registry to regenerate only changed prompts; covered 90% application modules.
- Built a Swagger/OpenAPI-driven **API test generation framework in Python** that auto-creates dependency-ordered CRUD suites, resolves runtime IDs, and reuses authenticated sessions across tenant/workspace contexts.
- Added structured reporting (Allure, HTML, JSON) and AI-based failure analysis to speed up debugging of automated test runs.
- Implemented UI enhancements on the company website using React, TypeScript, and Material UI.

Data Intern **May 2025 – Aug 2025**
Techment Technology, Bhilai *Tools Used: Microsoft Fabric, Azure, Power BI*

- Built **ETL pipelines in Microsoft Fabric** to ingest on-prem MySQL data into OneLake with incremental loads and schema validation, improving pipeline reliability by 40%.
- Optimized PySpark notebooks and Dataflow Gen2 pipelines, reducing Power BI refresh latency by 35%.

Projects

Swagger-Based API Test Automation Framework | *Python, pytest, Swagger, Allure, TypeScript, OpenAI SDK*

- Generated runnable pytest suites for Create/Get/Update/Delete flows directly from a Swagger file; **topologically sorted** create calls so parent entities exist before dependents.
- Built an ID-resolution engine that falls back from in-run memory to suggestion endpoints to list endpoints, and skips cleanly when no valid record exists, keeping tests stable without hardcoded IDs.
- Automated auth with a Playwright-based token capture and a refresh-before-expiry manager that injects bearer and tenant/account/workspace headers into every request.
- Shipped Allure, HTML, and PDF run reports plus an LLM analyzer that suggests root causes for failed tests.

End-to-End UI Test Automation Pipeline | *TypeScript, Playwright, Codex CLI, Playwright MCP, Node.js*

- Automated the full UI testing cycle with one command: refresh login session, detect new or changed feature prompts, generate Playwright specs, run the suite, and publish reports.
- Cut repetitive scripting by having an AI agent explore the live app and produce Playwright tests with real selectors, forms, and navigation for CRUD workflows across 90% of the modules.
- Added a **SHA-256 prompt registry** so only new or modified prompts are regenerated, and reused saved browser storage state to skip repeated logins.
- Generated HTML and JSON reports with screenshots and traces on failure, with per-run folders for easy debugging.

Education

B.Tech Computer Science & Engineering **Aug 2022 – July 2026**
Manipal University Jaipur, Jaipur *GPA: 8.9*